Question 1: LDA

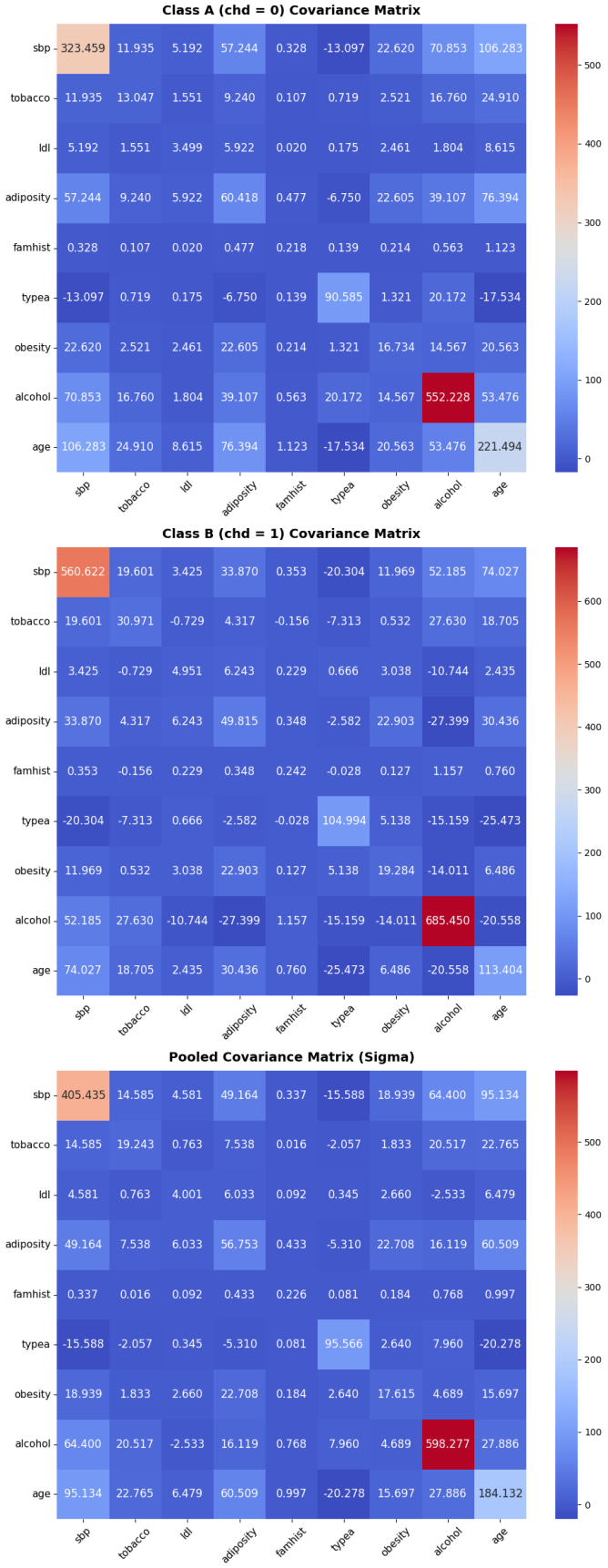
a. Calculate the prior probabilities, class means, and pooled covariance matrix from these data. Use the pooled covariance denominator $n - K$ from the lecture, where $K = 2$ is the number of classes.

Prior Probabilities

Class	Prior Probability
Class 0	0.6537
Class 1	0.3463

Class Means (Centroids)

Feature	Class 0 Mean	Class 1 Mean
sbp	135.460	143.738
tobacco	2.635	5.525
ldl	4.344	5.488
adiposity	23.969	28.120
famhist	1.318	1.600
typea	52.368	54.494
obesity	25.737	26.623
alcohol	15.931	19.145
age	38.854	50.294


Class A (chd = 0) Covariance Matrix
sbp 323.459 11.935 5.192 57.244 0.328 -13.097 22.620 70.853 106.283
tobacco 11.935 13.047 1.551 9.240 0.107 0.719 2.521 16.760 24.910
ldl 5.192 1.551 3.499 5.922 0.020 0.175 2.461 1.804 8.615
adiposity 57.244 9.240 5.922 60.418 0.477 -6.750 22.605 39.107 76.394
famhist 0.328 0.107 0.020 0.477 0.218 0.139 0.214 0.563 1.123
typea -13.097 0.719 0.175 -6.750 0.139 90.585 1.321 20.172 -17.534
obesity 22.620 2.521 2.461 22.605 0.214 1.321 16.734 14.567 20.563
alcohol 70.853 16.760 1.804 39.107 0.563 20.172 14.567 552.228 53.476
age 106.283 24.910 8.615 76.394 1.123 -17.534 20.563 53.476 221.494
sbp tobacco ldl adiposity famhist typea obesity alcohol age
500 400 300 200 100 0
Class B (chd = 1) Covariance Matrix
sbp 560.622 19.601 3.425 33.870 0.353 -20.304 11.969 52.185 74.027
tobacco 19.601 30.971 -0.729 4.317 -0.156 -7.313 0.532 27.630 18.705
ldl 3.425 -0.729 4.951 6.243 0.229 0.666 3.038 -10.744 2.435
adiposity 33.870 4.317 6.243 49.815 0.348 -2.582 22.903 -27.399 30.436
famhist 0.353 -0.156 0.229 0.348 0.242 -0.028 0.127 1.157 0.760
typea -20.304 -7.313 0.666 -2.582 -0.028 104.994 5.138 -15.159 -25.473
obesity 11.969 0.532 3.038 22.903 0.127 5.138 19.284 -14.011 6.486
alcohol 52.185 27.630 -10.744 -27.399 1.157 -15.159 -14.011 685.450 -20.558
age 74.027 18.705 2.435 30.436 0.760 -25.473 6.486 -20.558 113.404
sbp tobacco ldl adiposity famhist typea obesity alcohol age
600 500 400 300 200 100 0
Pooled Covariance Matrix (Sigma)
sbp 405.435 14.585 4.581 49.164 0.337 -15.588 18.939 64.400 95.134
tobacco 14.585 19.243 0.763 7.538 0.016 -2.057 1.833 20.517 22.765
ldl 4.581 0.763 4.001 6.033 0.092 0.345 2.660 -2.533 6.479
adiposity 49.164 7.538 6.033 56.753 0.433 -5.310 22.708 16.119 60.509
famhist 0.337 0.016 0.092 0.433 0.226 0.081 0.184 0.768 0.997
typea -15.588 -2.057 0.345 -5.310 0.081 95.566 2.640 7.960 -20.278
obesity 18.939 1.833 2.660 22.708 0.184 2.640 17.615 4.689 15.697
alcohol 64.400 20.517 -2.533 16.119 0.768 7.960 4.689 598.277 27.886
age 95.134 22.765 6.479 60.509 0.997 -20.278 15.697 27.886 184.132
sbp tobacco ldl adiposity famhist typea obesity alcohol age
500 400 300 200 100 0

b. Use your estimated values to predict the label of each observation in the training data. These are in-sample fitted labels. Provide the confusion matrix, with observed classes in rows and predicted classes in columns. Check that the dimensions of your matrices and vectors match. Calculate the scores yourself for this part.

Accuracy (258 + 87) / 462 = **74.7%**

Sample Predictions (Training Data)

Obs	Actual Class	Score (Class 0)	Score (Class 1)	Predicted Class	Correct?
0	1	64.9105	65.9311	1	✓
1	1	66.0667	65.1850	0	✗
2	0	54.6649	53.6643	0	✓

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	258 — True Negative (TN)	44 — False Positive (FP)
Actual 1	73 — False Negative (FN)	87 — True Positive (TP)

c. Perform the same LDA analysis using `MASS::lda` in R or `LinearDiscriminantAnalysis(solver="svd")` in Python, using the same numerical predictors and empirical class priors. Provide the confusion matrix. Do the results match?

Yes, same as the heatmap.

	sbp	tobacco	ldl	adiposity	famhist	typea	obesity	alcohol	age
sbp	405.43	14.59	4.58	49.16	0.34	-15.59	18.94	64.40	95.13
tobacco	14.59	19.24	0.76	7.54	0.02	-2.06	1.83	20.52	22.77
ldl	4.58	0.76	4.00	6.03	0.09	0.34	2.66	-2.53	6.48
adiposity	49.16	7.54	6.03	56.75	0.43	-5.31	22.71	16.12	60.51
famhist	0.34	0.02	0.09	0.43	0.23	0.08	0.18	0.77	1.00
typea	-15.59	-2.06	0.34	-5.31	0.08	95.57	2.64	7.96	-20.28
obesity	18.94	1.83	2.66	22.71	0.18	2.64	17.62	4.69	15.70
alcohol	64.40	20.52	-2.53	16.12	0.77	7.96	4.69	598.28	27.89
age	95.13	22.77	6.48	60.51	1.00	-20.28	15.70	27.89	184.13

```
lda_sk = LinearDiscriminantAnalysis(solver="svd", priors=[pi[0], pi[1]])

lda_sk.fit(X, y)

y_pred_sk = lda_sk.predict(X)

match = np.array_equal(y_pred_manual, y_pred_sk)
```

```
print(f"\nDo manual predictions match scikit-learn? {match}")

print("\n--- Pooled Covariance Matrix (Sigma) ---")

print(np.round(Sigma_pooled, 2))
```

Question 2: QDA and marginal screening

a. **Fit QDA using these 40 predictors and report the test confusion matrix, with observed classes in rows and predicted classes in columns. Also report the test misclassification rate.**

Diagonal (bold) = correct predictions. The biggest confusion is Class 4 being predicted as Class 9 (25 cases), and Class 3 misclassified as Class 5 and Class 8.

Test misclassification rate: 15.55%

	P0	P1	P2	P3	P4	P5	P6	P7	P8	P9
O0	**319**	0	24	6	2	2	1	0	5	0
O1	0	**238**	5	0	11	0	7	0	3	0
O2	8	0	**168**	8	8	3	0	0	3	0
O3	5	0	4	**128**	0	14	0	0	15	0
O4	0	2	13	1	**150**	1	0	1	7	25
O5	12	0	3	5	1	**131**	0	0	6	2
O6	0	0	8	0	1	8	**148**	0	5	0
O7	1	0	6	1	6	0	0	**124**	2	7
O8	1	0	10	12	3	2	0	0	**137**	1
O9	0	0	1	2	7	1	1	4	9	**152**

b. **Does the method work? Why do you think it works or does not work?**

It works because it resolves singularity: When using all 256 pixels, many pixels are constant or highly collinear within specific digit classes (e.g., corner pixels that are always blank for certain digits), making the within-class covariance matrices singular (non-invertible). Screening down to 40 pixels reduces dimensionality so that each class's covariance matrix can be successfully inverted without crashing.

c. **Repeat the analysis using the 10 predictors with the largest marginal variances in the training data. How does the test misclassification rate compare with the 40-predictor version? Why do you think this happens?**

Test misclassification rate: 34.23%

Much worse than 40 predictors. The biggest problem classes are Class 2 (only 95/198 correct), Class 4 (95/200, with 47 spilling into Class 9), and Class 6 (78/170, with 41 mistaken for Class 0).

Dropping from 40 predictors down to 10 makes the covariance matrices even more stable to estimate, but it usually increases the misclassification error rate. This happens because 10 pixels discard too much structural shape information about the handwritten digits, making it harder for QDA to distinguish between similar numbers (like a 3 and an 8).

	P0	P1	P2	P3	P4	P5	P6	P7	P8	P9
O0	**298**	0	11	1	1	16	28	4	0	0
O1	0	**241**	10	0	3	1	1	0	2	6
O2	19	2	**95**	14	14	3	22	21	5	3
O3	5	0	6	**103**	3	42	0	0	7	0
O4	1	5	14	0	**95**	2	4	15	17	47
O5	15	0	6	25	4	**96**	4	2	8	0
O6	41	0	15	4	0	18	**78**	7	5	2
O7	4	1	6	0	6	0	1	**122**	0	7
O8	1	2	11	36	12	11	4	1	**74**	14
O9	0	2	9	2	20	0	0	16	10	**118**

Question 3: Compare LDA and QDA

a. **Generate one dataset for each value of ρ. Fit both methods and calculate their test misclassification rates. Within each setting, use the same training and test observations for the two methods.**

Rho show correlation between the first two variables.

Rho = 0.0 | LDA Test Error: 0.1505 | QDA Test Error: 0.1925

Rho = 0.2 | LDA Test Error: 0.1820 | QDA Test Error: 0.2040

Rho = 0.4 | LDA Test Error: 0.1715 | QDA Test Error: 0.1790

Rho = 0.6 | LDA Test Error: 0.1705 | QDA Test Error: 0.1775

Rho = 0.8 | LDA Test Error: 0.1635 | QDA Test Error: 0.1350

Rho = 0.9 | LDA Test Error: 0.1565 | QDA Test Error: 0.1000

b. **At each fixed value of ρ, repeat the experiment 50 times, generating new training and test observations in each repetition. Plot average test error against ρin one figure, with one curve for LDA and one for QDA. These curves compare performance across different data-generating distributions.**

Rho show correlation between the first two variables.

Rho = 0.0 | Mean LDA Error: 0.1717 | Mean QDA Error: 0.2004

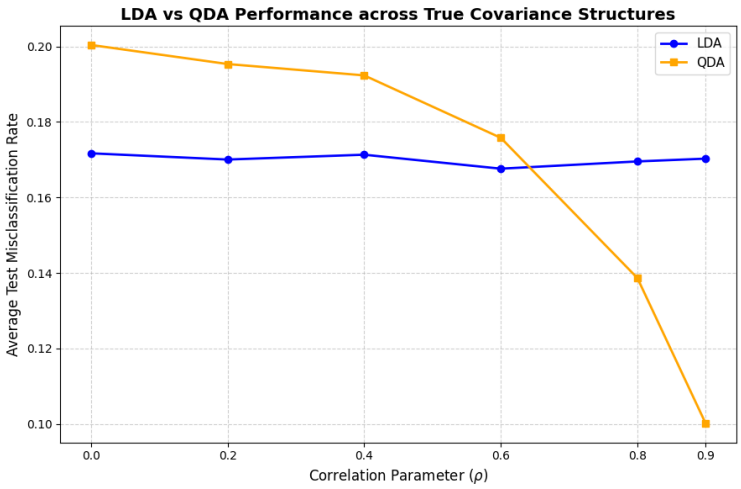
Rho = 0.2 | Mean LDA Error: 0.1700 | Mean QDA Error: 0.1953

Rho = 0.4 | Mean LDA Error: 0.1713 | Mean QDA Error: 0.1923

Rho = 0.6 | Mean LDA Error: 0.1676 | Mean QDA Error: 0.1758

Rho = 0.8 | Mean LDA Error: 0.1695 | Mean QDA Error: 0.1387

Rho = 0.9 | Mean LDA Error: 0.1703 | Mean QDA Error: 0.1001



c. Compare the two methods at small and large values of ρ. How does changing the true covariance structure affect their relative performance? Explain the pattern using their covariance assumptions and the bias-variance trade-off. How can the eight noise covariates affect performance even though they contain no information about the class label?

Property	LDA	QDA
Class means	Separate means	Separate means
Covariance	One shared covariance matrix	One covariance matrix per class
Decision boundary	Linear	Generally quadratic
Model complexity	Lower	Higher
Main advantage	More stable estimation	Can capture class-specific covariance

1. LDA, assumes the classes share the same covariance structure: higher bias, lower variance.

Its shared-covariance assumption can be wrong when the classes have different correlations. However, it estimates fewer parameters, making it less sensitive to random variation in the training sample.

2. QDA, allows each class to have its own: lower bias, higher variance.

It can represent the true class-specific covariance structures, but estimating two full covariance matrices from finite samples introduces more uncertainty.

Although X3-X10 contain no information about the class label, the models do not automatically know that. They must estimate their means, variances, and covariance relationships from the training sample. Random fluctuations can make these estimated relationships appear useful even though they are not truly predictive.

This can increase estimation variance and worsen test performance. QDA is particularly vulnerable because it estimates separate covariance matrices for the two classes. The important distinction is: A variable can contain no true class information but still hurt a fitted model by introducing unnecessary parameters and estimation noise.

Question 4: Create and use an LDA/QDA analysis skill

Invocation prompt used:

"Apply my LDA/QDA skill to the dataset at /Users/runting/stat432-fall2026/homework/week-07/data/saheart.csv. Random seed: 4."

Skill explanation (paste this into your homework):

The skill file SKILL.md (4,889 bytes) is stored outside automatically loaded skill folders and is only loaded on explicit request. The skill is dataset-agnostic: it accepts only a file path and an optional random seed at invocation. The agent then inspects the data independently — determining the response variable, identifying and dummy-coding categorical predictors, and summarizing class distributions — before running the full LDA/QDA analysis. This design makes the skill reusable on any tabular classification dataset, with no dataset-specific variable names or values appearing in the skill itself.

SKILL: LDA/QDA Classification Analysis

Invocation
Load this skill only when explicitly requested with a phrase such as
"apply my LDA/QDA skill" or "invoke the LDA/QDA skill." Do not load
automatically. The user supplies only the dataset file path at invocation
time. The agent inspects the data and determines the response variable
and predictors independently before running the analysis.

Step 1: Data Exploration

Before any modeling, read the dataset and report:

1. **Dimensions**: number of rows and columns.
2. **Variable types**: identify which columns are numeric, binary, or
   categorical (character/factor).
3. **Response variable**: identify the most likely classification target
   — typically a binary or low-cardinality integer/factor column. If
   ambiguous, choose the column that appears last or is most clearly
   a label. State your choice and the reason.
4. **Predictors**: use all remaining columns as predictors. Convert any
   character or factor columns to numeric dummy variables (0/1). Note
   which columns were converted and how.
5. **Class distribution**: report the count and proportion of each class.
   Note any imbalance.
6. **Summary statistics**: report mean and standard deviation of each

numeric predictor.

All decisions made in this step must be stated clearly before any modeling begins.

Step 2: Preprocessing

1. Dummy-code all character or factor predictors using training-fold data only; apply the same encoding to validation and test folds.
2. Standardize all predictors (zero mean, unit variance) using training-fold statistics. Apply the same scaling to validation/test folds.
3. For PCA variants, retain enough principal components to explain at least 90% of training-fold variance. Record the number of components kept for each fold.

Step 3: Candidate Models

Fit all four candidates and compare via cross-validation:

ID	Model
M1	LDA on original predictors
M2	QDA on original predictors
M3	PCA → LDA (≥90% variance retained)
M4	PCA → QDA (≥90% variance retained)

All preprocessing is estimated inside each CV fold using training observations only.

Step 4: Data Split

- Use the random seed supplied at invocation. If none is supplied, use seed 42 and state this.
- Hold out 20% of observations as a **test set** before any modeling.
- Use the remaining 80% as the **training set** for all CV and model selection steps.
- Never use the test set until the final evaluation of the selected model.

Step 5: Cross-Validation Procedure

- Apply **5-fold cross-validation** on the training set.
- Use the **same fold assignments** for all four candidates.
- Metric: **misclassification rate** (proportion misclassified in each held-out fold).
- Report mean CV misclassification rate ± standard deviation across folds for each candidate.
- Select the candidate with the lowest mean CV misclassification rate.

Break ties by preferring the simpler model (M1 > M3 > M2 > M4).

Step 6: Final Model and Test Evaluation

- Refit the selected model (and all preprocessing) on all training observations.
- Evaluate the fitted model **once** on the reserved test set.
- Report the test misclassification rate as the estimated prediction error. Do not revisit or re-tune after seeing the test result.

Step 7: Required Report (≤ 1 printed page)

Produce a concise report with exactly the following sections:

1. Data and Candidate Models
One short paragraph describing the dataset (dimensions, response, predictors, any coding applied), the four candidate models, and the CV strategy (folds, metric, seed).

2. Cross-Validation Results (Table)
A table with columns: Model, Mean CV Error, SD CV Error.
Bold or mark the selected model's row.

3. CV Error Comparison (Figure)
A bar chart or dot plot showing mean CV misclassification rate for each candidate, with error bars (±1 SD). Label axes clearly. Include the random seed in the caption.

4. Selected Model and Test Performance
One paragraph stating which model was selected, why (CV results), and the test misclassification rate. Include the confusion matrix on the test set.

5. Findings and Limitations
Two to four sentences interpreting the result: which classes are hardest to separate, whether PCA helped or hurt, and any notable limitations (sample size, class imbalance, linearity assumption).

Constraints

- Use only LDA, QDA, or their combination with PCA.
- No variable names from any specific dataset may appear in this skill.
- The report must fit on one printed page; the confusion matrix may be compact (numbers only).
- State the random seed explicitly in the report.
- The skill is dataset-agnostic: it applies to any tabular classification dataset, not only the dataset used at time of writing.

LDA/QDA Analysis of Coronary Heart Disease Risk — SAheart Data

seed = 4 · n = 462 · train = 369 · test = 93 · 5-fold CV · response: chd (binary)

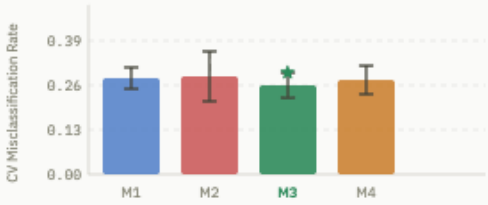
CANDIDATE MODELS

Four classifiers were compared: plain LDA (M1) and QDA (M2) on all nine original predictors, and PCA-preprocessed variants LDA (M3) and QDA (M4) where predictors were standardized and reduced to principal components explaining at least 90% of training-fold variance. The character predictor *famhist* was dummy-coded (Present = 1, Absent = 0) before fitting. All scaling and PCA were estimated within each CV fold using training observations only and applied to the held-out fold without refitting.

CROSS-VALIDATION RESULTS

MODEL	MEAN CV ERR	SD
M1: LDA	0.2846	0.0331
M2: QDA	0.2872	0.0716
M3: PCA-LDA ★	**0.2629**	**0.0387**
M4: PCA-QDA	0.2818	0.0427

★ Selected model. 7 PCs retained (93.2% variance).



Mean 5-fold CV error ± 1 SD. Seed = 4. ★ = selected model.

SELECTED MODEL & TEST PERFORMANCE

19.35%
Test misclassification rate

7 PCs
Components retained (≥90% var.)

	Pred 0	Pred 1
Obs 0	57	4
Obs 1	14	18

PCA-LDA (M3) was selected based on the lowest 5-fold CV misclassification rate (26.29%), outperforming plain LDA (28.46%), PCA-QDA (28.18%), and QDA (28.72%). The model was refit on all 369 training observations using 7 principal components (93.2% of variance explained), then evaluated once on the held-out test set of 93 observations, yielding a test misclassification rate of 19.35% (18 errors out of 93).

FINDINGS & LIMITATIONS

PCA preprocessing modestly improved LDA performance, suggesting mild multicollinearity among the nine predictors (sbp, obesity, and adiposity are likely co-linear). The classifier struggles most with true CHD cases: of 32 CHD-positive test observations, 14 (44%) were misclassified as negative, indicating low sensitivity. QDA showed higher variance across folds (SD = 0.072) than LDA variants, reflecting instability from estimating separate covariance matrices on a modest training set. The linearity assumption of LDA may not fully capture non-linear risk boundaries in cardiovascular data; tree-based or logistic methods may improve sensitivity at modest cost to specificity.